

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>Book</u>		
Property Name	Data Type	Source Tag
<u>BookID</u>	<u>int</u>	<u>System Generated</u>
<u>Title</u>	<u>string</u>	<u>User Input</u>
<u>Author</u>	<u>string</u>	<u>User Input</u>
<u>Genre</u>	<u>string</u>	<u>User Input</u>
<u>TotalCopies</u>	<u>int</u>	<u>User Input</u>
<u>IsAvailableCopies</u>	<u>bool</u>	<u>Default Value</u>
<u>BorrowCount</u>	<u>int</u>	<u>Default Value</u>

Entity Name: <u>Member</u>		
Property Name	Data Type	Source Tag
<u>memberId</u>	<u>int</u>	<u>System Generated</u>
<u>memberName</u>	<u>string</u>	<u>User Input</u>
<u>phoneNumber</u>	<u>string</u>	<u>User Input</u>
<u>email</u>	<u>string</u>	<u>User Input</u>
<u>address</u>	<u>string</u>	<u>User Input</u>
<u>registrationDate</u>	<u>string</u>	<u>Default Value</u>
<u>status</u>	<u>string</u>	<u>Default Value</u>

Entity Name: <u>Loan</u>		
Property Name	Data Type	Source Tag
<u>loanId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>From List</u>
<u>bookId</u>	<u>int</u>	<u>From List</u>
<u>borrowDate</u>	<u>string</u>	<u>Default Value</u>
<u>dueDate</u>	<u>string</u>	<u>Calculated</u>
<u>returnDate</u>	<u>string</u>	<u>User Input</u>
<u>fineAmount</u>	<u>double</u>	<u>Calculated</u>

Entity Name: <u>Reservation</u>		
Property Name	Data Type	Source Tag
<u>reservationId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>From List</u>
<u>bookId</u>	<u>int</u>	<u>From List</u>
<u>reservationDate</u>	<u>string</u>	<u>Default Value</u>
<u>notificationSent</u>	<u>bool</u>	<u>Default Value</u>
<u>status</u>	<u>string</u>	<u>Default Value</u>
<u>queuePosition</u>	<u>int</u>	<u>Calculated</u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

Write the relationship cycle here:

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterMember	Creates a new library member
2	AddBook	Adds a new book to the library
3	UpdateBook	Updates book information
4	BorrowBook	Creates a loan and decreases available copies
5	ReturnBook	Records return and updates stock
6	CalculateFine	Calculates overdue fines
7	ReserveBook	Creates a reservation for unavailable books
8	NotifyMember	Notifies members when reserved books become available
9	ViewActiveLoans	Displays all active loans
10	MostBorrowedBooksReport	Shows books ranked by borrow count

step 2:

A member must first register before using the library.

When a member borrows a book, a Loan record is created. The member is selected from the member list and the book is selected from the available books list. The availableCopies value of the selected book decreases by one and the borrowCount increases by one. When the member returns the book, the Loan record is updated with the return date. If the return date is after the due date, the system calculates a fine amount. The book's availableCopies value increases by one.

If all copies of a book are currently borrowed, a member can create a Reservation. A Reservation record is created and linked to both the member and the book.

When a reserved book becomes available, the system notifies the next member in the reservation queue and updates the reservation status.

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
<code>guestId</code>	int	System Generated
<code>guestName</code>	string	User Input
<code>guestPhone</code>	string	User Input
<code>guestEmail</code>	string	User Input
<code>guestNationality</code>	string	User Input
<code>registrationDate</code>	string	Default Value

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	System Generated
roomNumber	string	User Input
roomType	string	User Input
floorNumber	int	User Input
nightlyPrice	decimal	User Input
isAvailable	bool	Default Value

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	System Generated
guestId	int	From List
roomId	int	From List
checkInDate	string	User Input
checkOutDate	string	User Input
numberOfNights	int	Calculated
totalPrice	decimal	Calculated
status	string	Default Value

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	System Generated
bookingId	int	From List
guestId	int	Calculated
roomId	int	Calculated
actualCheckIn	string	Default Value
actualCheckOut	string	User Input
finalAmount	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	RegisterGuest	Creates a new guest
2	AddRoom	Adds a room to the hotel
3	ViewAvailableRooms	Displays available rooms
4	CreateBooking	Creates a room booking
5	CancelBooking	Cancels a booking
6	CheckInGuest	Creates a check-in record
7	CheckOutGuest	Completes guest checkout

#	Service Name	What it does
8	CalculateStayCost	Calculates total stay cost
9	RoomAvailabilityReport	Shows room availability status

Domain 2:

A guest first registers in the hotel system. A Guest record is created.

The guest can then make a room booking by selecting an available room. A Booking record is created and linked to both the Guest and Room entities. The selected room's availability changes to false. When the guest arrives, a CheckInRecord is created using the booking information. The booking, guest, and room details are copied into the check-in record.

The total price of the booking is calculated from the room's nightly rate multiplied by the number of nights. When the guest checks out, the actual check-out date is recorded and the final amount is calculated. The room becomes available again for future bookings.

If a booking is cancelled before check-in, the booking status changes to Cancelled and the room immediately becomes available again.